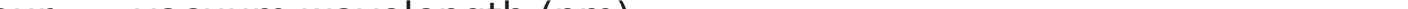
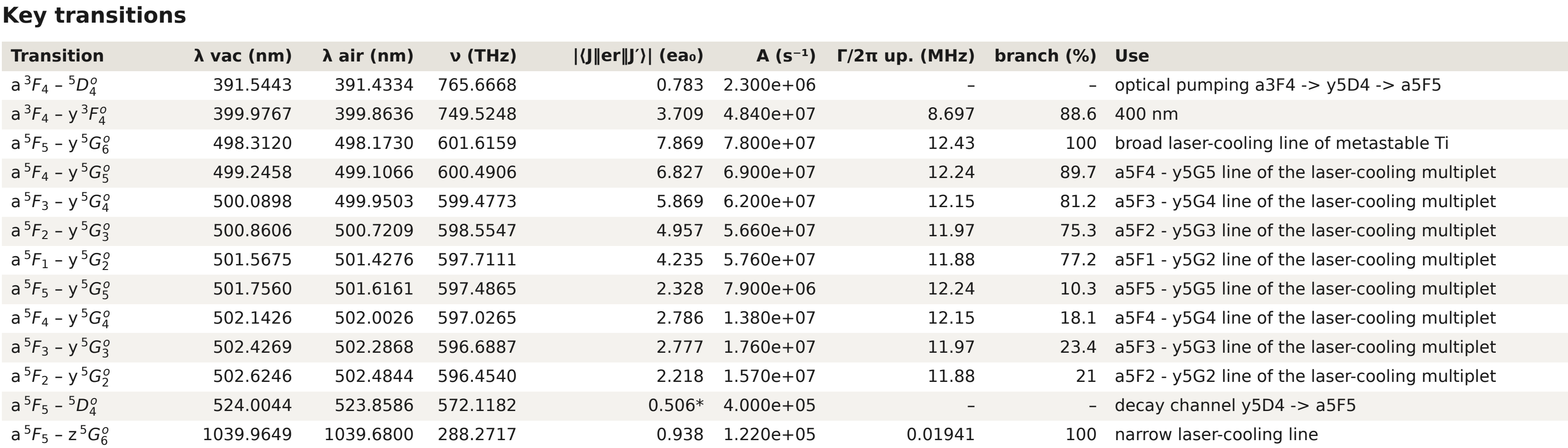


Ti⁺ ionisation limit 55072.50 cm⁻¹ = 6.82812 eV (181.579 nm)

Arrow colour = vacuum wavelength (nm)



300 400 500 600 700 800 1000 1200 1500 2000



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (mHz)									
a_1^F	-52.904(7)	28.082(9)	F=1/2: +545.90	F=3/2: +458.96	F=5/2: +316.87	F=7/2: +123.84	F=9/2: -114.23	F=11/2: -389.76				
$a_1^F a_1$	-37.9918(6)	37.681(10)	F=3/2: +493.40	F=5/2: +387.32	F=7/2: +243.04	F=9/2: +64.81	F=11/2: -141.92	F=13/2: -370.50				
$a_1^F a_2$	-74.8450(25)	-28.67(5)	F=5/2: +1110.25	F=7/2: +856.99	F=9/2: +527.93	F=11/2: +120.49	F=13/2: -368.49	F=15/2: -942.73				
$z_1^F a_0^0$	-85(9)	77(5)	F=1/2: +1518.12	F=9/2: +1112.00	F=11/2: +625.25	F=13/2: +63.65	F=15/2: -365.98	F=17/2: -1255.75				
$y_0^F a_0^0$	-96.60(20)	-0(1)	F=3/2: +1207.38	F=5/2: +965.95	F=7/2: +627.92	F=9/2: +193.27	F=11/2: -338.04	F=13/2: -966.06				
$y_0^F d_0^0$	-14.2(3)	-0(7)	F=1/2: +248.06	F=3/2: +184.50	F=11/2: +106.67	F=13/2: +145.51	F=15/2: -92.11	F=17/2: -213.28				

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$a^1F_u - {}^3D_u^o$	391.544	46-48	-943(3)
$a^1F_u - {}^3D_u^o$	391.544	50-48	884(3)
$a^1F_u - {}^3D_u^o$	391.544	47-48	-448(3)
$a^1F_u - {}^3D_u^o$	391.544	49-48	474(3)
$a^1F_u - y^1G_g$	498.312	46-48	-716.2(4)
$a^1F_u - y^1G_g$	498.312	50-48	671.6(21)
$a^1F_u - y^1G_g$	498.312	47-48	-346(3)
$a^1F_u - y^1G_g$	498.312	49-48	355(3)
$a^1F_u - y^1G_g$	499.246	46-48	-706.7(9)
$a^1F_u - y^1G_g$	499.246	50-48	703.2(12)
$a^1F_u - y^1G_g$	500.090	46-48	-703.2(12)
$a^1F_u - y^1G_g$	500.090	50-48	660.1(6)
$a^1F_u - y^1G_g$	500.861	46-48	-691.3(20)
$a^1F_u - y^1G_g$	500.861	50-48	652.0(5)
$a^1F_u - y^1G_g$	501.567	46-48	-682.7(3)
$a^1F_u - y^1G_g$	501.567	50-48	642.1(4)
$a^1F_u - y^1G_g$	501.756	50-48	-714.0(6)
$a^1F_u - y^1G_g$	501.756	50-48	682.9(6)
$a^1F_u - y^1G_g$	502.143	46-48	-702(5)
$a^1F_u - y^1G_g$	502.143	50-48	670.6(11)
$a^1F_u - y^1G_g$	502.427	46-48	-690.1(5)
$a^1F_u - y^1G_g$	502.427	50-48	660.29(28)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$